

Leveraging Deep Q-Learning to maximize consumer quality of experience in smart grid

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ARTICLE INFO

Keywords:

Smart grid

Deep Q-Learning

Real-time pricing

Quality of experience

ABSTRACT

The smart grid system has addressed the problems of traditional power grid by not only meeting energy demand in real-time but also limiting its wastage. The **two key objectives** of a smart grid system are to ensure a higher Quality of Experience (QoE) for consumers and to **reduce consumer costs** using dynamically varying pricing concepts. However, these two objective parameters oppose each other as maximizing the consumer QoE requires the availability of sufficient electric power at any given time, which in turn increases power purchase cost. In this paper, we introduce an efficient power management system architecture of a smart grid and develop an Optimal Energy Allocation and Prediction system based on Deep Q-Learning, namely OEAP-DQL, that brings a trade-off between the two. The developed OEAP-DQL system is a multi-objective linear programming (MOLP) problem that predicts consumer electricity demand by exploiting four different weighted and regressive moving average forecasting methods in the action space to accurately capture dynamically varying customer demand behaviors. Furthermore, iterative exploitation of multiple learning methods decreases forecasting error and intelligent stored power management helps the OEAP-DQL smart grid operator (SGO) to enhance its profit. The results of our simulation experiments show that the OEAP-DQL system outperforms the state-of-the-art works in terms of QoE and cost.

1. Introduction

In the present era, we find ourselves immersed in a contemporary society that is progressively embracing digital advancements. In our digital life, we are being dependent on IoT devices and to survive our life, we are feeling starving of electrical energy. Smart grid technology enables the two-way flow of electricity and information between power providers and consumers [1]. It uses sensors, automation, and communication technologies to collect and analyze data in real-time, which allows power providers to better manage the supply and demand of electricity. This helps to reduce energy waste and prevent blackouts, which can save money and improve the reliability of the power grid [2]. Moreover, smart grid technology facilitates the incorporation of renewable energy sources like solar and wind power into the power grid. This is particularly significant since harnessing renewable energy not only aids in reducing carbon emissions but also contributes to minimizing fuel costs within the optimal power flow problem, thereby playing a crucial role in mitigating the global warming impact associated with

power generation [3]. The smart grid plays a pivotal role in this scenario by enabling power providers to efficiently manage the integration of renewable energy sources into the power grid. This, in turn, ensures a dependable and stable supply of electricity [4,5]. According to a report by the U.S. Department of Energy, the smart grid technology could reduce electricity consumption by up to 4% and also save consumers \$20 billion annually by 2030 [6].

However, the smart grid presents a wide range of challenges for researchers, industry professionals, and policymakers [7]. These challenges include optimizing the profitability of service providers, enhancing the quality of service and overall consumer experience, boosting consumer satisfaction or utility, and mitigating consumer power costs [8]. Furthermore, challenges persist on the SGO's side with the implementation of smart metering systems and power storage devices. The integration and management of these technologies pose additional hurdles that require innovative solutions. Effectively tackling these challenges is pivotal for the successful deployment and operation

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<https://doi.org/10.1016/j.energy.2023.130165>

Received 7 August 2023; Received in revised form 18 December 2023; Accepted 27 December 2023

Available online 2 January 2024

0360-5442/© 2023 Published by Elsevier Ltd.

of smart grid infrastructures, ensuring reliability, sustainability, and optimal performance [9]. Furthermore, the complexity of smart grid systems is escalating with the growing global population and the proportional increase in consumer needs. With the growth of population and industries, managing the electricity demand has become a difficult task for the grids [10]. Similarly, when there is a decrease in the number of people utilizing electric power, the demands on the power grid are reduced accordingly. Therefore, the development of advanced algorithms and models becomes imperative to optimize grid operations, handle the variability and intermittency of renewable energy sources, and accurately predict electricity loads in specific areas [11]. Additionally, one of the primary hurdles in these situations is to maximize the quality of experience (QoE) for the consumer while also minimizing their power consumption costs.

A lot of related work has been done to address the challenges of smart grid systems; however, there is a lot of uncertainty about the power distribution process in the existing literature. The authors in [12] have developed an optimal dynamic pricing mechanism to bring a trade-off between consumer utility and operator profit in smart grid systems without considering the consumer power cost. Most importantly, they have not considered any power storage where the extra power can be stored to support when a power shortage occurs. Another group of authors in [13] have developed an artificial intelligence-based dynamic pricing and demand response algorithm. However, they have considered fixed-rate or time-of-use (ToU) pricing schemes. Furthermore, a significant challenge involves improving consumer satisfaction levels while simultaneously reducing electricity costs. In [14], the authors have developed a consumer-aware pricing policy model, where they have tried to minimize customer cost by using a genetic algorithm (GA). However, they consider neither consumer satisfaction nor service provider profit. These observations have motivated us to develop a framework that can effectively allocate electricity and forecast consumer power demand in real-time. The major research questions that are answered in this paper can be listed as follows,

- A. How can we allocate the required electricity to consumers to maximize their satisfaction and minimize the power purchase cost?
- B. Which prediction model can enhance the accuracy of electricity demand prediction at different time slots?
- C. What amount of performance improvement can be achieved in terms of consumer QoE and power purchase cost by exploiting the developed electricity allocation and prediction model?

In this paper, we have presented an Optimal Electricity Allocation and Prediction model exploiting Deep Q-Learning namely OEAP-DQL. The developed OEAP-DQL system establishes a trade-off between two opposing aims in smart grid infrastructure i.e., increasing customer QoE and reducing consumer power cost. The problem of implementing the two aforementioned objectives is expressed using a multi-objective linear programming (MOLP) problem. When the consumers take service from SGO at a lower cost, then the rate of QoE is high inversely. However, if the consumer's QoE is high that means consumers take continuous service from SGO, which increases the consumer's power cost. In this scenario, we have introduced the real-time pricing (RTP) scheme which is updated hourly based on the consumer's power demand. As a result, consumer power demand plays a vital role in this smart grid system. Therefore, we have also established an electricity demand prediction algorithm based on Deep Q-Learning (DQL) to forecast the need for electrical power in light of historical data, real-time data, and prevailing weather patterns. The developed OEAP-DQL system highlights key implications and applications, including enhanced grid reliability, energy consumption optimization by integrating renewable energy sources, real-time pricing, power storage system, and smart meter incorporation, which helps to promote the sustainable and cost-effective system. The main contributions of this paper are summarized as follows,

- We have designed an efficient power management system architecture for a smart grid and developed an Optimal Energy Allocation and Prediction system based on Deep Q-Learning (OEAP-DQL).
- The OEAP-DQL has been formulated as a multi-objective linear programming (MOLP), which has made a trade-off between QoE and the power cost of consumers based on dynamic real-time electricity pricing within the consumer budget.
- We have built an electricity prediction model using DQL, leveraging historical data to predict power demand. This DQL-driven forecasting model incorporates various forecasting models within the action space to adapt to dynamic consumer demands, resulting in a reduction in forecasting errors.
- We have demonstrated the efficacy of our developed OEAP-DQL system and the simulation results show that the proposed system outperforms the state-of-the-art works in terms of QoE and consumers power cost as high as 10% and 15%, respectively.

The rest of this work is structured as follows: Section 2 describes the existing literature on smart grid systems. In Section 3, we present the system model and assumption. Subsequently, Section 4 describes the detailed design issues of the developed OEAP-DQL system. After that, the performance analysis result and comparison with the state-of-the-art works to validate the developed system is discussed in Section 5. Finally, the conclusion is presented in Section 6.

2. Related work

The smart grid field encompasses a broad range of research areas and it is rapidly evolving by researchers and industry professionals around the world. Numerous research works have been identified by experts in the smart grid like demand response, energy storage, power electronics, renewable energy integration, micro-grids, and so on [15–18].

The authors in [19] have investigated a distributed online algorithm for electricity distribution in smart grid environments, which considers several factors such as consumer's utility, grid load smoothing, and energy provisioning cost while also protecting consumer's privacy. However, the operator's profit cannot be determined, and they also overlook the dynamic fluctuations of power prices over time. R. Lu et al. [13] have presented a hierarchical electricity market model with mathematical formulations and a reinforcement learning (RL) methodology for optimizing energy management under changing ambient conditions. However, the main limitation is that the proposed algorithm's performance is inferior when measured the performance against the established methods, such as fixed-rate or time-of-use (ToU) schemes. In addition, they have not considered any mechanism to enhance consumer satisfaction levels while also lowering electricity costs. Another group of authors in [12] have proposed an optimal dynamic pricing mechanism for smart grid systems namely ODPT, which balances consumer utility and operator profit. However, the authors have not considered that external factors, such as weather, and economic conditions, can affect power consumption. Additionally, they have not taken into account the usage of power storage to store surplus energy and provide electrical support during power shortages. The authors in [20] have proposed a real-time incentive-based demand response scheme for smart grids using RL and deep neural networks. The developed scheme optimizes customer incentive rates based on individual needs and preferences, leading to improved system performance and potential economic benefits for the service provider. However, they have neglected the consideration of consumer power costs and dissatisfaction levels.

A general framework for modeling electricity retail pricing based on load demand and market price information, to minimize system cost while integrating renewable energy sources is presented in [21]. However, the impact of consumer behavior and preferences on electricity

Table 1
Comparison of OEAP-DQL with other models.

Model	Consumer QoE	Consumer cost	Load prediction	Consumer budget	SGO profit	Real-time pricing
DOA [19]	✓	X	X	X	✓	X
DPR [13]	X	✓	✓	X	X	X
ODPT [12]	X	X	✓	X	✓	X
CAPP [14]	X	✓	X	✓	X	X
DRL-MG [28]	✓	✓	X	X	X	✓
OEAP-DQL	✓	✓	✓	✓	✓	✓

consumption has been overlooked, which can affect the effectiveness of the proposed pricing policy. In [22], the authors have proposed an energy broker model to facilitate energy trading between the existing power grid and energy consumers using renewable energy certificates. However, the authors have not considered the impact of uncertainties in energy generation and demand on the proposed energy trading model. Additionally, the paper assumes that all energy consumers have access to an energy storage system (ESS), which may not be realistic. B. Mohandes et. [23] have discussed the need for operating reserves in modern power grid systems due to the increasing penetration of variable renewable energy sources and proposed the deployment of Demand Response (DR) and ESS to provide operating reserves. It also presents a multi-objective optimization problem to maximize grid revenue and minimize consumer energy costs. However, the feasibility of the proposed solutions may depend on factors such as technology availability, cost, regulatory framework, and consumer behavior.

A day-ahead hourly-based pricing scheme for end consumers in the smart home community architecture has been developed in [24], which minimizes the difference between day-ahead and real-time prices to benefit the end consumer by decreasing their consumption cost. However, the proposed model assumes that all consumers will be able to adhere to the suggested scheduling provided by the controller, which may not be realistic. Another group of authors in [25] present a deep reinforcement learning (DRL) approach for optimizing multi-energy systems in smart grids. They formulate the optimal control problem of production and storage units as a partially observable Markov decision process (POMDP) and propose an actor-critic DRL algorithm to solve it. The proposed approach is tested on a novel multi-energy residential micro-grid model, and further investigation is needed to assess its suitability for larger-scale multi-energy smart grids such as eco-districts and smart cities. In [26], the authors have proposed a multiple load demand response strategy to guide consumer's energy consumption habits and participate in the coordinated operation of the integrated energy system. They have established a two-stage collaborative optimization model and demonstrate that considering the demand response of electricity load can improve the overall economic benefit of the system. However, the proposed model and demand response strategy require significant changes in the existing energy system infrastructure and consumer behavior, which could be challenging to implement in practice. Another group of authors in [27] have introduced Deep4Ener, a single deep learning model designed for forecasting energy demand across multiple consumers. However, the evaluation is conducted on limited data. Another group of authors in [14] has developed a model called the consumer-aware pricing policy (CAPP), which utilizes a genetic algorithm (GA) to minimize customer costs. However, their model neglects both consumer satisfaction and service provider profits. Additionally, they have utilized a fixed rate pricing policy, known as the ToU rate scheme.

The study in [29] explores the application of RL and recurrent neural networks for heating, ventilation, and air-conditioning (HVAC) systems optimization in data centers, considering real-time pricing and temperature signals. However, the paper acknowledges the challenge of assuming a Markov property, which may not be suitable for cost optimization under real-time pricing due to the time-dependent reward function. A real-time energy management system using deep

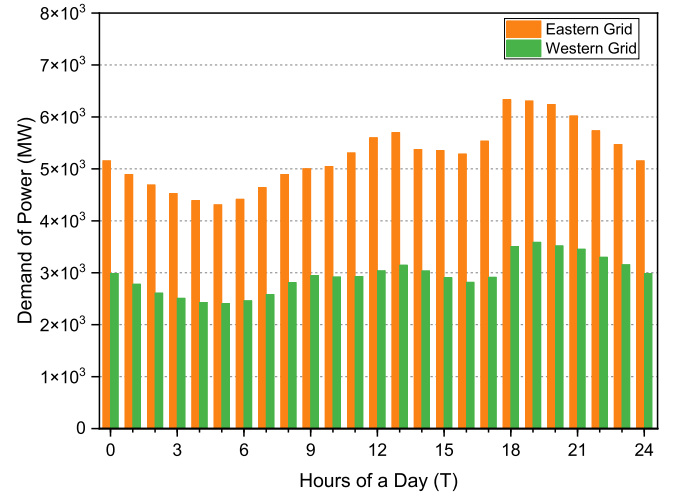


Fig. 1. Electricity consumption in the different periods.

reinforcement learning, namely DRL-MG to improve customer service and lower costs is introduced in [28]. However, it lacks analysis on DRL-MG's scalability in larger microgrids and does not address trade-offs with factors like provider profit and sustainability in scheduling electricity demand. Sheha et al. [30] have used real-time electricity prices as decision variables to incentivize consumer investments in energy storage and reduce costs. Various machine learning techniques are exploited to predict citywide energy demand and the nonlinear auto-regressive neural network shows the best accuracy. However, the paper does not address the grid's complexity, the need for data-driven economic models, scalability to different areas, or consumer behavior's impact on variable prices and potential price manipulation. Table 1 summarizes the comparative studies of the state-of-the-art works.

Most of the existing works predominantly focus on enhancing consumer QoE or maximizing SGO profit, often neglecting the impact on consumer power costs in the smart grid infrastructure. However, these two factors are inherently contradictory, as maximizing consumer QoE ensures the availability of electric power at all times, but it also leads to higher consumer power costs. Conversely, minimizing consumer power costs requires reducing power purchases from the SGO, which may compromise consumer QoE. Moreover, forecasting the precise power demand in future time slots is a critical factor in enhancing consumer's QoE. Recognizing these dynamics, we have been motivated to develop a comprehensive framework that predicts the future power demand by exploiting Deep Q-Learning and facilitates trade-offs between the minimization of consumer power costs and maximizing QoE in the smart grid infrastructure.

3. System model and assumptions

This section introduces a novel computation framework of the proposed model, the power demand scheme of the developed system architecture, and the assumptions.

3.1. Demand scheme

The consumer's electricity consumption for a whole day long for two different areas which are- (i) the Eastern grid area and (ii) the Western grid area is depicted in Fig. 1. These simulation data were collected from the Power Grid Company of Bangladesh (PGCB) [31]. In our proposed system, electricity demand is divided into three different categories based on different demand levels which are - Low-peak, Mid-peak, and High-peak. We have considered 1 AM to 8 AM to be the

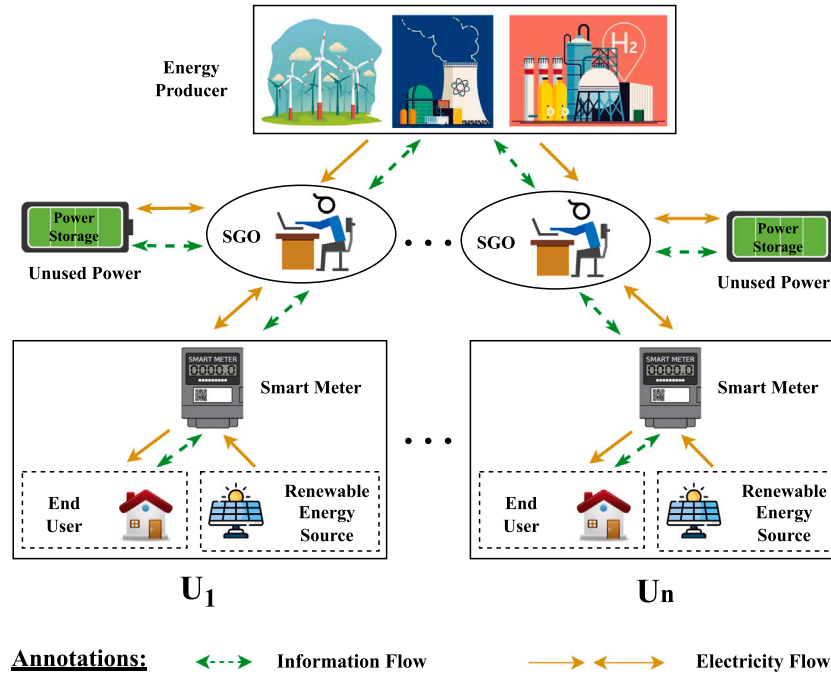


Fig. 2. Smart grid power allocation management system model.

low-peak region, 9 AM to 10 AM, and 1 PM to 2 PM to be the mid-peak region, and the rest of them are in the high-peak region with respect to Eastern Grid and Western Grid. According to the consumer's power demand, we are considering three different price schemes based on real-time pricing (RTP), which is shown in Eq. (1). Here, selling price Y^{hp} is for high-peak regions, Y^{mp} is for mid-peak regions, and selling price Y^{lp} is for low-peak regions.

$$Y(t) = \begin{cases} Y^{lp}(t), & \text{for low-peak region;} \\ Y^{mp}(t), & \text{for mid-peak region;} \\ Y^{hp}(t), & \text{for high-peak region.} \end{cases} \quad (1)$$

In most cases, the electricity price changes depending on the electricity demand, which is revised every hour using smart meter data based on load demand profiles. This data is then used to determine the current level of electricity demand and to adjust the price accordingly. To maintain a consistent supply of electricity during peak demand periods, it is crucial to exchange information between the SGOs and the consumers in real-time. Therefore, a unique cost function involving both the current level of electricity demand and the cost of generating and delivering that electricity is necessary.

3.2. Smart grid system environment

In our proposed system, there are three components: consumer, SGO, and energy producer (EP). The set of electricity consumers, SGOs, and EPs are denoted by U , N , and M , respectively. The system components i.e., consumer $i \in U$, SGO $k \in N$, and EP is $j \in M$ communicate with each other according to Fig. 2. Electricity is consumed by consumers in various sectors, including residential, commercial, and industrial. The EP is largely in charge of electricity generation and delivery to the SGOs. To fulfill its objectives, the SGO communicates information among consumers as well as EPs and thereby controls the general operations of the entire area. The total timeline is divided into equal time slots $t \in T$ and the SGO purchases electricity from EP under wholesale market rates and sells it to customers at retail market rates based on the consumer demand at each time slot $t \in T$. To precisely predict the consumer's electricity usage over time, we have

used the power demand estimating model, which helps SGOs distribute electric power effectively. We introduce a vital smart metering system for advanced monitoring, which facilitates the transfer of consumer power demand patterns to the SGOs and exchange the information at various time slots, and allows the SGOs to accurately monitor and analyze the consumption patterns of consumers, enabling them to make informed decisions regarding power management and distribution and control in smart grids. Smart meters are typically accessed by end users or consumers through the energy service provider's interface and are managed by SGOs for overall grid management [32]. We also emphasize that individuals can get benefit from renewable energy sources like solar panels, and wind turbines due to the limitation of energy. Users or consumers can harvest energy during shortages, save costs, and contribute to the national grid [33]. consumers send power demand to the SGO by subtracting the power generated from renewable sources from the overall power demand of consumers, referred to as the consumer's actual power demand. For those without renewable sources, the overall power demand aligns with the consumer's actual power demand. In this work, we assume that the consumers who can access smart meters and who can utilize renewable energy sources and contribute to the smart grid system can benefit from the developed system. The major notations that have been used throughout the whole paper are listed in Table 2.

4. Design of OEAP-DQL system

In this section, we first design an operational system architecture that introduces the functional components at pre-entities of a smart grid system and their interrelationships made in the OEAP-DQL system. Then, we unfold an optimization framework for optimal energy allocation by considering the consumer's QoE and cost. For efficiently allocating power, we develop a Deep Q-Learning (DQL) based high accuracy electricity demand prediction model. We conclude this section by presenting the power storage system employed at each SGO and providing an illustrative example for a better operational process understating of the OEAP-DQL.

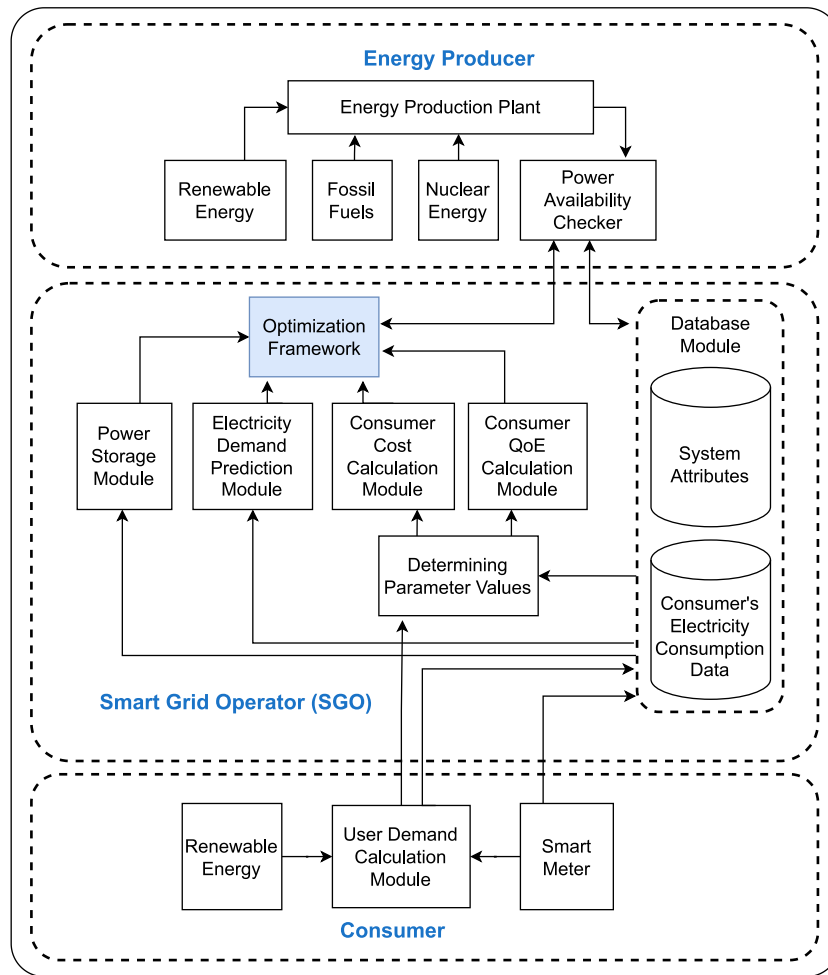


Fig. 3. System architecture of the proposed OEAP-DQL system.

Table 2

List of notations.

Notation	Description
U	Set of all power consumers in the system
M	Set of all EP in the system
N	Set of SGO in the system
$a_i(t)$	Allocated power to consumer i at time slot t
$e_i(t)$	Electricity demand consumer i at time slot t
$p_j(t)$	Per unit purchase cost by SGO from EP j at time slot t
$C(t)$	Consumer cost of consumer at time slot t
$E_d(t)$	Total predicted demand from all consumer's at time slot t
$E_e(t)$	Actual electricity demand from all consumer's at time slot t
$Y(t)$	Per unit selling price at time slot t
$Q_i(t)$	Consumer satisfaction level at time slot t
$\mu_k^s(t)$	Total store power of power storage at SGO at time slot t
$\mu_k^{sp}(t)$	Total shortage power of SGO at time slot t
$B(t)$	Budget of consumer at time slot t
$P_{SGO}(t)$	SGO profit at time slot t
$A_j(t)$	Available power at EP $j \in M$ at time slot t

4.1. System architecture

The operational system architecture of the developed OEAP-DQL is illustrated in Fig. 3 comprising three integral components: the Consumer, Smart Grid Operator (SGO), and Energy Producer.

- **Consumer Modules:** Initially, the consumer accesses electricity generated from renewable sources, aligning with sustainable practices. In cases where additional electricity is required, users can

seamlessly communicate their demand to the SGO through the Demand Calculator Module. Real-time electricity consumption data and integration with the main grid are facilitated by smart meters, enhancing the efficiency of the system.

- **Database Module:** It stores Consumer's Electricity Consumption Data and System Attributes data for different computational parameters.
- **Determining Parameter Values:** This module is responsible for collecting relevant parameters from system attributes, consumer electricity consumption database, and user demand calculator module and propagates data for calculation of consumer power purchase cost and QoE.
- **Power Storage Module:** This module manages the storage and distribution of excess power, ensuring optimal utilization and reduced wastage.
- **Electricity Demand Prediction Module:** This module predicts future electricity demands based on consumer electricity consumption historical data exploiting the DQL technique, enabling proactive adjustments to the energy distribution.
- **Consumer Cost Calculation Module:** It optimizes energy distribution strategies to minimize costs for consumers, fostering economic efficiency.
- **Consumer QoE Calculation Module:** This module focuses on enhancing the QoE for consumers by optimizing energy delivery to meet their preferences and needs.

- **Energy Producer Modules:** The energy producer responds to consumer demands communicated by the SGO, adapting its production to meet dynamic user needs. They maintain the energy production plant, which produces energy from renewable energy sources, fossil fuels, and nuclear energy sources. They contribute to a sustainable energy ecosystem by providing the necessary power while maintaining a harmonious balance with the environment.

4.2. Optimization framework

Our first objective is to maximize the QoE of consumers, whereas the second objective aims to minimize consumer costs. The aforementioned two conflicting objectives can be formulated using Multi-Objective Linear Programming (MOLP) as follows,

Maximize:

$$\mathcal{O} = \sum_{i \in U} \sum_{t \in T} \{ \delta \times Q_i(t) - (1 - \delta) \times \hat{C}_i(t) \}, \quad (2)$$

where, $Q_i(t) \in [0, 1]$ is the QoE of the consumer's, which also indicates the consumer satisfaction level at time slot t , that can be calculated by using Eq. (8). $\hat{C}_i(t) \in [0, 1]$ is the value of normalized consumer cost at time slot t , which is calculated by using Eq. (10) and $\delta \in [0, 1]$ is a weight factor whose value can be tuned to control the objective function. If $\delta = 1$, then it makes Eq. (2) a consumer QoE maximization problem. On the other hand, $\delta = 0$ converts it to a consumer cost minimization problem, while any other values of δ make the desired trade-off between consumer QoE and consumer cost. Eq. (2) is subject to the following constraints:

Power allocation constraint,

$$0 \leq a_i(t) \leq e_i(t), \quad \forall i \in U, \forall t \in T \quad (3)$$

Power purchase constraint,

$$E_{d,k}(t) \leq A_j(t), \quad \forall j \in M, \forall t \in T, \forall k \in N \quad (4)$$

Profit constraint,

$$Y_k(t) > \frac{\sum_{j \in M} \{ E_{d,j}(t) \times p_j(t) \}}{\sum_{j \in M} E_{d,j}(t)} \quad (5)$$

Power demand constraint,

$$\sum_{i \in U} (a_i(t) \times \eta_i) \leq \sum_{j=1}^m E_{d,k}(t), \quad \forall t \in T, \forall k \in N \quad (6)$$

Consumer budget constraint,

$$B_i(t) \geq C_i(t), \quad \forall i \in U, \forall t \in T \quad (7)$$

Here, Eq. (3) denotes the power allocation constraint that ensures for a consumer $i \in U$, the amount of allocated power must be less than or equal to consumer electricity demand $e_i(t)$. Eq. (4) is the power purchase constraint, which means at time slot t , SGO $k \in N$ cannot purchase more power from any EP $j \in M$ than its available power $A_j(t)$. The profit constraint in Eq. (5) guarantees that, at time slot t , the per-unit selling price $Y_k(t)$ of SGO $k \in N$ should be higher than its purchase cost. SGO dynamically adjusts the upper bound per unit price p_j , adhering to real-time pricing theory, to maximize its profit. Eq. (6) is the power demand constraint which means at time slot t , total consumer allocated power must be less than or equal to total predicted demand. Eq. (7) is a consumer budget constraint which means that for a consumer $i \in U$, the total consumer budget must be greater than or equal to the consumer cost of electricity service, which can be determined using Eq. (9).

4.2.1. Calculation of QoE

Consumer QoE refers to the satisfaction or enjoyment that consumers derive from using a product or service. In this work, we quantify consumer QoE as the ratio of allocated power to the consumer over the consumer's power demand. Tracking QoE metrics can be a valuable tool for optimizing power allocation and improving the overall consumer experience of power distribution systems. The consumer's QoE at time slot t can be determined as follows,

$$Q_i(t) = \frac{a_i(t)}{e_i(t)} \times \eta_i, \quad (8)$$

where, $e_i(t)$ is the power demand of consumer $i \in U$ and $a_i(t)$ is the allocated power to consumer $i \in U$ at time slot t . η depicts a binary decision variable. If consumer $i \in U$ takes service from SGO, then $\eta_i = 1$ otherwise, $\eta_i = 0$. Here, $Q_i(t)$ always provides a normalized value within the interval $[0, 1]$. The value of $Q_i(t)$ is proportional to the satisfaction level of consumer $i \in U$ at time slot t . A higher value of $Q_i(t)$ indicates that the consumer i can enjoy the expected power demand, increasing its satisfaction.

4.2.2. Calculation of consumer cost

Electrical consumption data is traded between customers and utilities, who promptly adjust their pricing structures. In the OEAP-DQL system, consumer electricity consumption is observed and the new per-unit rate is enforced through the use of dynamic pricing scheduling and intelligent smart meters. Through the communication network infrastructure, smart meters and SGO can share information, such as figuring out how much energy is needed and exchanging data (like selling and wholesale prices). The control center will estimate the real-time price policy for all grid consumers based on load fluctuations in smart grid processes. When consumer demand approaches the upper limits of supplied electricity, the per-unit electricity price Y has been increased immediately and vice versa. Electricity purchase cost C_i of consumer $i \in U$ at time slot t can be estimated as follows,

$$C_i(t) = \eta_i \times a_i(t) \times Y(t), \quad (9)$$

where, $t \in T$ denotes a billing period of 24 equally spaced time slots. The normalized consumer cost $\hat{C}$ can be calculated as:

$$\hat{C}_i(t) = \frac{C_i(t)}{B_i(t)}, \quad (10)$$

where, $\hat{C}_i(t) \in [0, 1]$ and $B_i(t)$ denotes the budget of consumer $i \in U$ at time slot t .

4.3. DQL based electricity demand prediction model

To make a better trade-off between consumer's QoE and cost, we need to know the power demand of all consumers. Therefore, power demand plays a vital role in the OEAP-DQL system. Power demand prediction models, particularly in load forecasting, face uncertainty arising from several factors like dynamic energy consumption, weather variability, renewable energy integration, random events, and human factors. Given the uncertain nature of power demand influenced by these factors, accurately predicting electricity demand in future time slots becomes challenging. If we can predict the power demand of consumers accurately, the developed system gives a more accurate result. For that, we establish a Deep Q-Learning (DQL) based power demand prediction model. The DQL model often resembles the Markov Decision Process (MDP). In Q-learning, the probabilities of going from one state to another are not explicitly defined in the model. Instead, the model must learn this, as the agent traverses the environment repeatedly to learn the best strategy by itself. Therefore, Q-learning is suitable in those scenarios where specific probabilities, rewards, and penalties are not completely known. Now, here we will explain the major components of the DQL-based prediction model in the following subsections.

4.3.1. State space

In the proposed OEAP-DQL system, the whole day is divided into equal time slots. We have considered total time slot T and each time slot is considered as a state s_t . Therefore, the state space can be represented as follows,

$$S_t = \{s_1, s_2, \dots, s_T\}, \quad 1 \leq t \leq T. \quad (11)$$

4.3.2. Action space

In each state, the agents will make a decision or take an action $a_i \in A$ on what to do next depending on the current system state. In our developed model, we have considered four actions to forecast the electricity demand in the future time slots, which are **Weighted Average (WA)** model, **Exponentially Weighted Moving Average (EWMA)** model, **Auto Regressive Integrated Moving Average (ARIMA)** model, and **Seasonal Auto-Regressive Integrated Moving Average (SARIMA)** model.

- (i) **Weighted Average (WA)**: In action a_1 , the agent uses a weighted average technique of the electricity demand in different states $s_t \in S$ to predict the optimal consumer demand. Whenever the electricity demand is stable, this method works most effectively [34]. Among all of the previous states $s_t \in S$, the agent provides more weight to the most recent states and less weight to the former states. Therefore, the WA model-based action can forecast electricity demand at time slot t as follows,

$$E_d(t+1) = \frac{\sum_{i=1}^T \omega_i \times E_d(t)}{\sum_{i=1}^T \omega_i}. \quad (12)$$

The weight value of a particular state is represented by ω_i , and the electrical demand value of that state is represented by $E_d(t)$.

- (ii) **Exponentially Weighted Moving Average (EWMA)**: In action a_2 , the agent predicts the electricity demand of different SGOs of a particular state through the EWMA technique. The exponential increasing/decreasing rule is applied to predict energy demand in each slot. The predicted electricity demand using the EWMA technique can be determined as follows,

$$E_d(t+1) = \beta \times E'_d(t) + (1 - \beta) \times E_d(t), \quad (13)$$

where, $E_d(t)$ and $E'_d(t)$ denote the predicted electricity demand and the actual electricity demand at time slot t , respectively. The value of the weighting factor is defined as $\beta = 0.125$, which is set by the extensive simulation.

- (iii) **Auto Regressive Integrated Moving Average (ARIMA)**: In action a_3 , the agent predicts the electricity demand of different SGOs in a particular state, through the ARIMA model. The ARIMA model is a linear regression model that utilizes historical data and estimates electricity consumption for each time interval. The Auto-Regressive (AR) model predicts electricity demand by comparing the electricity demand of the current time slot to the electricity demand of prior time slots. The Moving Average (MA) model calculates the difference in error between the current and prior time slots [35]. Combining the AR and MA models yields the ARIMA model, which examines variations in electricity demand and error between the prior and present time intervals for decision-making. It can predicate anticipated power demand $E_d(t)$ at time slot t as follows,

$$E_d(t) = c + \beta_1 E_d(t-1) + \beta_2 E_d(t-2) + \dots + \beta_p E_d(t-p) \kappa(t) + \phi_1 \kappa(t-1) + \phi_2 \kappa(t-2) + \dots + \phi_q \kappa(t-q), \quad (14)$$

where, c represents the estimated intercept term, and κ defines the forecast error. β and ϕ are the weighting factors for the AR and MA models, respectively; whereas, p and q indicate the order of these models, respectively. The ARIMA model is proficient in analyzing time series data with stationary behavior, effectively capturing dependencies, and trends. It proves valuable for

short to medium-term forecasting when predictable patterns are present, but its performance may be less optimal for data with significant changes or complex interactions over time. Moreover, it does not support seasonal data.

- (iv) **Seasonal Auto-Regressive Integrated Moving Average (SARIMA)**: In action a_4 , the agent predicts the electricity demand of different SGO of a particular state, through the Seasonal ARIMA or SARIMA model, which is an extension of ARIMA that explicitly supports uni-variate time series data with a seasonal component. While the ARIMA model is excellent overall, it can be very challenging to incorporate seasonality and exogenous variables into the model. Since the ARIMA model implies a stationary time series; therefore, we have utilized the SARIMA model. SARIMA model incorporates both seasonal as well as non-seasonal elements [36]. The model can be defined as,

$$ARIMA(p, d, q) \times (P, D, Q)_\zeta$$

Here, p, d, q denote the non-seasonal AR order, non-seasonal differences, and non-seasonal MA order, respectively. On the other hand, P, D, Q indicate the seasonal AR order, seasonal differences, and seasonal MA order, respectively. ζ is the repetitive seasonal pattern's period. The non-seasonal elements can be represented as follows:

$$\begin{aligned} AR : \phi(E_d) &= 1 - \phi_1 E_d - \dots - \phi_p E_d^p \\ MA : \theta(E_d) &= 1 + \theta_1 E_d + \dots + \theta_q E_d^q \end{aligned} \quad (15)$$

The seasonal elements are:

$$\begin{aligned} AR : \Phi(E_D^s) &= 1 - \Phi_1 E_D^s - \dots - \Phi_P E_D^{Ps} \\ MA : \Theta(E_D^s) &= 1 + \Theta_1 E_D^s + \dots + \Theta_Q E_D^{Qs} \end{aligned} \quad (16)$$

By Combining Eq. (15) and (16), the SARIMA model can be formally expressed as follows,

$$\phi(E_d) \Phi(E_D^s) Y_t = \theta(E_d) \Theta(E_D^s) \psi_t, \quad (17)$$

where, Y_t represents the ordinary and seasonal difference in electricity demand, considering non-seasonal differences, d and seasonal differences, D . The Gaussian white noise term, ψ_t accounts for the unpredictable and random elements in the time series, while Y_t captures the variations in electricity demand. Specifically, within the SARIMA framework, ϕ signifies the auto-regressive parameters governing the non-seasonal component, and θ represents the moving average parameters for the non-seasonal aspect. Additionally, Φ and Θ denote the auto-regressive and moving average parameters, respectively for the seasonal component. These parameters serve as coefficients that establish the connection between the current observation and past observations in the time series. Their role is crucial in capturing both non-seasonal and seasonal auto-correlation and moving average patterns within the data.

4.3.3. Reward

At each iteration, the agent receives a reward $\mathcal{R}$ for successfully performing the correct action. In the OEAP-DQL system, we have taken the reward value as the ratio of the real electricity demand E_r of a particular state and predicted electricity demand E_d from the Deep Q-Learning algorithm. Then the reward value can be defined as follows,

$$\mathcal{R} = \begin{cases} \frac{E_r}{E_d}, & P^e(t) \leq P_{max}^e \\ -1, & \text{otherwise} \end{cases} \quad (18)$$

Here, $P^e(t)$ and P_{max}^e denote the error rate at time slot t and the maximum allowable error rate, respectively.

Algorithm 1 DQL-based Electricity Demand Prediction Algorithm

Input: Number of episodes E , State Space S , Action Space A
Output: Approximate Q-value for each state-action pair

- 1: Initialize replay memory M with capacity D
- 2: Initialize the training network with a set of weight θ
- 3: Initialize the target network with a set of weight θ'
- 4: Initialize the exploration rate ϵ
- 5: **for** Episode=1, E **do**
- 6: Reset the environment with state $s_t \in S$
- 7: **for** $t=1, T$ **do**
- 8: Observe current state s_t
- 9: **if** Probability $\geq \epsilon$ **then**
- 10: Select an action $a_t \in A$
- 11: **else**
- 12: Select $a_t = \arg \max_{a_t \in A} Q^*(s_t, a_t; \theta)$
- 13: **end if**
- 14: Execute action a_t and observe reward R_t from Eq. (18)
- 15: Store (s_t, a_t, R_t, s_{t+1}) into Q
- 16: Random-minibatch d' from replay memory M
- 17: **for all** $d_i \in d'$ **do**
- 18: Calculate $\mathcal{Z}$ by using Eq. (19)
- 19: Calculate $L_{(w)}$ by using Eq. (21)
- 20: Perform Gradient Descent on $L_{(w)}$ with respect to θ
- 21: **end for**
- 22: Update the next state s_{t+1}
- 23: **end for**
- 24: **end for**
- 25: **return** Q-value for each state-action pair

4.3.4. Training of DQL algorithm

In our method, both the state and action space have a lot of dimensions. Because of this, it is hard to get all $Q(s_t, a_t)$ and put them in the Q-table. However, the reinforcement learning approach of OEAP-DQL uses a separate neural network with weight vector θ' to find out the target Q-values. The targeted or expected state-action Q-value $\mathcal{Z}$ from the Q table for taking an action $a_t \in A$ from state s_t in Deep Q-Learning by exploiting Bellman's equation can be expressed as follows,

$$\mathcal{Z} = R_{(s_t, a_t)} + \gamma \times \max_{a_t \in A} \hat{Q}(S_{t+1}, a_{t+1}; \theta'). \quad (19)$$

Here, the discount factor γ can be any number in the range ($0 \leq \gamma \leq 1$). When γ is close to 0, the agent prioritizes short-term reward, whereas when it is closer to 1, the agent also considers the potential for long-term reward. Where, the Q-value $Q(s_t, a_t)$ can be updated as follows,

$$Q(s_t, a_t) = Q(s_t, a_t) + \alpha [r(s_t, a_t) + \gamma \times \max_{a_t \in A} \hat{Q}(S_{t+1}, a_{t+1}) - Q(s_t, a_t)] \quad (20)$$

In the DQN-based system, we want to minimize the loss function value $L_{(w)}$ between the target and predicted Q-value. The loss function value $L_{(w)}$ using the mean squared loss function can be determined as follows,

$$L_{(w)} = \{\mathcal{Z} - Q(s_t, a_t; \theta)\}^2. \quad (21)$$

The value of the loss function can be minimized with the help of the latest weight parameters θ , which DQL updates. We can obtain the gradient of the loss function defined in Eq. (21) using stochastic gradient descent similar to [37] by differentiating it with respect to θ as follows,

$$\theta := \theta - \alpha \frac{\partial}{\partial \theta} L_{(w)}. \quad (22)$$

In this scenario, α is the learning rate range in $0 \leq \alpha \leq 1$. By performing this stochastic gradient descent, it updates the θ value. The

process of updating the weights of the training network to minimize the loss function is accomplished through the application of Gradient Descent.

The training process of the DQL-based demand prediction algorithm is outlined in Algorithm 1. Initially, the replay memory M is initialized, along with the training network having random weights denoted as θ , and the target network with weight parameters θ' set equal to θ from line 1 to 3. We incorporate the experience replay technique with DQL due to the instability issue of DQL. When the DNN learns from sequential samples, the learning process becomes ineffective since there is a chance that the data will have a strong correlation. However, the experience replay technique can employ a single experience for several weight updates to enhance data efficiency [38]. In each episode t , the algorithm selects a new action a_t , using an ϵ -greedy policy. It balances exploration and exploitation by either randomly choosing an action from the action set, A with ϵ probability or doing a forward pass of the DQN to select the action a_t , with the maximum Q-value based on the state s_t . The chosen action is sent to the environment, which calculates a reward and serializes the next state, s_{t+1} . This experience, represented as the quadruplet (s_t, a_t, R_t, s_{t+1}) , is stored in the replay memory M . A minibatch is then sampled from the replay memory M , and the target network calculates targets for each state in the minibatch. The DQN is trained using minibatch gradient descent, updating weight parameters θ , based on the target Q-value and loss function. After each state update, s_{t+1} , the training network is cloned into the target network by setting θ' equal to θ . This process iterates till the maximum iteration number, T to prevent infinite training and ensure the stability and convergence of the DQN training process.

4.4. Managing power storage and allocation for SGOs using OEAP-DQL system

Consumer power demand cannot be estimated accurately in advance by SGO since it is private and very time-variable, especially at different prices. After getting the estimated power demand from the DQN model, SGO purchases the predicated electricity amount from the EPs and announces the prices to the consumers. To ensure an appropriate purchase quantity, an accurate estimation model is required. Two cases might occur after getting an estimated power demand.

- **Case-1:** In this case, if the OEAP-DQL system predicts power more than the consumer need, that is $E_{d,k}(t) > \sum_{i \in U} e_i(t)$, then the extra power will be stored at power storage. The extra power will be calculated as follows,

$$\mu_k^s(t) = E_{d,k}(t) - \sum_{i \in U} e_i(t) + \mu_k^s(t-1), \quad (23)$$

where, $\mu_k^s(t)$ is the total stored power of power storage at SGOs $k \in N$ at time slot t , $E_{d,k}(t)$ is the total predicted power demand by OEAP-DQL for SGO k , and $e_i(t)$ is the actual power demand of the consumers at time slot t .

Since, in the proposed model, we have considered power storage devices at the SGOs side, and for that reason, there is no power wastage occurring although OEAP-DQL predicts more power than the actual total power demand.

- **Case-2:** In this case, if the developed OEAP-DQL system predicts less power demand than the consumers need, that is $E_{d,k}(t) < \sum_{i \in U} e_i(t)$, then the SGOs will be using the power from power storage. On the other hand, if power storage has insufficient power then a power shortage occurs for a certain time.

$$\mu_k^{sp}(t) = \sum_{i \in U} e_i(t) - E_{d,k}(t) \quad (24)$$

$$\sum_{i \in U} a_i(t) = \begin{cases} E_{d,k}(t) + \mu_k^{sp}(t), & \mu_k^{sp}(t) \leq \mu_k^s(t) \\ E_{d,k}(t) + \mu_k^s(t), & \mu_k^{sp}(t) > \mu_k^s(t) \end{cases} \quad (25)$$

where, $\mu_k^{sp}(t)$ and $\mu_k^s(t)$ denote the total shortage and stored electricity power of SGOs at time slot t , respectively.

Table 3

Power usage, prediction and allocation statistics for one SGO.

Slot	Consumer demand (MW)	OEAP-DQL (MW)	Extra power (MW)	Stored power (MW)	Allocated power (MW)
0	5158.121	5261.283	+103.16	103.16	5158.12
1	4893.848	4942.786	+48.94	152.1	4893.85
2	4691.909	4551.149	-140.76	11.34	4691.91
3	4529.727	4642.97	+113.24	124.58	4529.73
4	4389.97	4389.97	0	124.58	4389.97
5	4311.348	4261.008	-50.34	74.24	4311.35
6	4417.318	4340.628	-76.69	0	4414.87
7	4640.455	4733.264	+92.81	92.81	4640.46
8	4895.818	4797.902	-97.916	0	4890.71
9	5005.379	5055.535	+50.16	50.16	5005.38
10	5050.667	5151.68	+101.01	151.17	5050.67
11	5311.242	5245.68	-65.562	85.61	5311.24
12	5602.545	5490.494	-112.051	0	5576.1
13	5700	5814	+114	114	5700
14	5378.424	5432.208	+53.78	167.78	5378.42
15	5354.864	5475.51	+120.65	288.43	5354.86
16	5287.788	5208.175	-79.613	208.82	5287.79
17	5536.788	5702.892	+166.1	374.92	5536.79
18	6336.227	6114.459	-221.768	153.15	6336.23
19	6308.379	6334.547	+26.17	179.32	6308.38
20	6244.939	6432.287	+187.35	366.67	6244.94
21	6021.379	5900.951	-120.428	246.24	6021.38
22	5736.015	5678.655	-57.36	188.88	5736.02
23	5471.348	5526.061	+54.71	243.59	5471.35

In this scenario, the proposed model enables SGOs to obtain substantial profits as they can sell all the power procured from the EPs. Additionally, power storage can mitigate power shortage issues.

4.5. An illustrative example

As previously stated, the predicted power demand can be higher than the actual demand or may significantly less than the actual demand from consumers during a particular time slot t , resulting in either of the following two scenarios, Case-1: OEAP-DQL system predicts more power than consumers need ($E_{d,k}(t) > \sum_{i \in U} e_i(t)$) or Case-2: OEAP-DQL system predict less power than consumers need ($E_{d,k}(t) < \sum_{i \in U} e_i(t)$). The following Table 3 shows the numerical illustration of the power usage and allocation for one SGO on different time slots.

In Table 3, at time slot zero, we can see that the consumer demand is 5158.121 MW and the predicted power is 5261.283 MW. As predicted power is greater than the consumer demand so this situation matches case 1. In this time slot, extra power is +103.16MW. For this time slot, the total allocated power will be 5158.12 MW, and the remaining power 103.16 MW will be stored in the power storage. After that at time slot 1, the consumer demand is 4893.848 MW and the predicted power is 4942.786 MW. In this case, the predicted power is again greater than the consumer demand so here also extra power will be stored. In time slot 1, extra power is 48.94 MW and the total stored power will be 152.1 MW. Here, the allocated power will be 4893.85 MW, as per consumer demand. After that, in time slot 2, the consumer demand is 4691.909 MW and the predicted power is 4551.149 MW. Here, the predicted power is less than the consumer demand, so this situation matches with case 2. Therefore, the extra power will be -140.76 MW and the allocated power is 4691.91 MW, as consumers demand. As the predicted power is less than the consumer demand, therefore, the remaining power will be delivered from the power storage. Therefore, the stored power will be decreased, and after the time slot 2, the stored power will be 11.34 MW.

From the Table 3 data, it is clear that the developed OEAP-DQL system reduces power wastage. If it predicts more electricity demand than the consumer's demand, the extra power will be stored in SGO's power storage, and it will be used in the next time slots. On the other hand, if the OEAP-DQL model predicts less consumer demand, then it

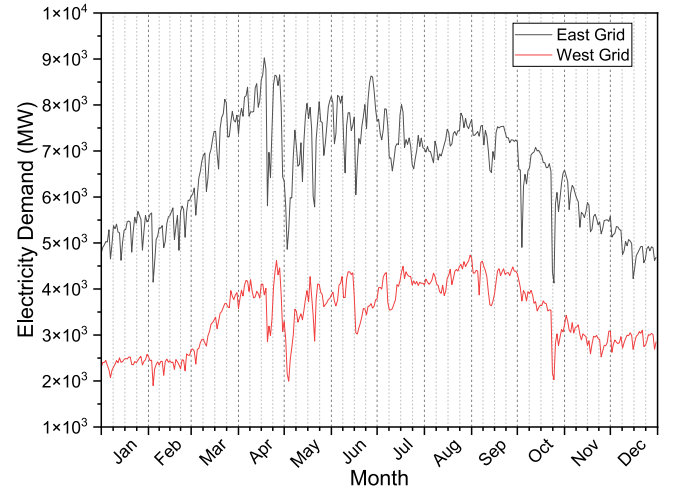


Fig. 4. Electricity load distribution of Dhaka city from January 2022 to December 2022 according to PGCB [31].

will deliver as per the consumer's demand from its power storage if the power storage has sufficient power. Therefore, we can conclude that the proposed OEAP-DQL system reduces power wastage and will maximize the consumer's QoE.

5. Performance evaluation

In this section, we have assessed the proposed OEAP-DQL system for predicting electricity demand and compared its performance with other state-of-the-art methods ODPT [12], DPR [13], CAPP [14] in terms of Quality of Experience, normalized consumer power cost, consumer satisfaction, and SGO profit. All algorithms and other result parameters we have implemented using Python programming version 3.10.2 in Google Colab.

5.1. Simulation environment

For the experimental purpose, we have collected the simulation data, and parameters from the Power Grid Company of Bangladesh (PGCB) [31]. This dataset covers one year of data, enabling us to analyze power-related trends, consumer behavior, and other relevant factors over an extended period. To illustrate the effectiveness and accuracy of our proposed method, we have chosen data recorded from January 2022 to December 2022 in Dhaka city, as depicted in Fig. 4. As shown in the figure, we can clearly observe that there is lower electricity demand from November to March due to the winter season in Bangladesh, which leads to reduced electric load on the grid during these months. In contrast, the electric load is higher in other months, particularly when Bangladesh experiences very hot weather. In the West Grid, from November to March, peak electricity demand ranges from 1899.12 MW to 3963.48 MW, whereas, the standard deviation of demand in this period is 208.50 MW. From April to October electricity demand ranged from 1995.76 MW to 4736.48 MW, and the standard deviation was 379.77 MW. On the other hand, For the East Grid from November to March, electricity demand ranged from 4148.84 MW to 8125.52; whereas, the standard deviation was 412.31 MW. From April to October, peak demand reached to 9024.24 MW, the low was 4128.32 MW; whereas, the standard deviation was 562.02 MW. However, in our model, daily operations were evaluated by analyzing power consumption over a 24-h task epoch similar to Fig. 5. Based on consumer participation, the 24-h time slots are divided into low-peak, mid-peak, and high-peak periods. The low-peak region spanned from 1 AM to 8 AM, the mid-peak region covered 9 AM to 10 AM and 1

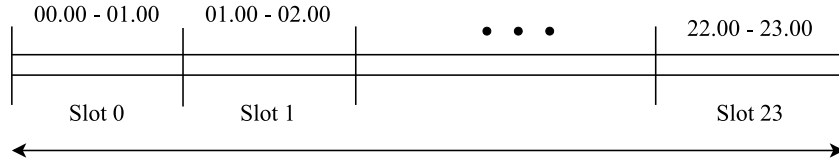


Fig. 5. A day representing 24 slots (00.00–23.00).

Table 4
Simulation parameters.

Parameter	Value
Simulation area	2000 × 2000 m ²
Number of consumers	100–1000
The demand of each consumer at each time slot	0–10 kW
Power generated from RESs of each consumer	0–6 kW
Per unit purchase price from EP at each time slot	\$5.0–\$12.0
Per unit selling price to consumers at each time slot	\$6.0–\$16.0
Electricity demand in low-peak range	[4200, 5500] KW
Electricity demand mid-peak range	[5500, 7000] KW
Electricity demand high-peak range	[7000, 1000] KW
Replay memory size (M) of DQN	10,000
Minibatch size	32
Learning Rate (α) of DQN	0.01
Turning Probability (ϵ)	0.1
Discount Factor (γ)	0.9
Number of Iterations for DQN	400

PM to 2 PM, and the remaining slots were categorized as high-peak, as depicted in Fig. 1.

For simulation purposes, we have considered a simulation model representing a power distribution system similar to the PGCB dataset serving an area with 100 to 1000 consumers. The actual power demand of each consumer is assumed to vary between 0–10 kW using uniform random distribution. The power generated from renewable energy sources (RESs) of each consumer is assumed to vary from 0–6 kW. The per unit purchase price of SGOs from the EPs is chosen randomly from \$5 to \$12 and the per unit selling price set by the SGOs ranges from \$6 to \$16 based on real-time pricing (RTP) schemes, which depended on the level of consumer participation. The user budget was randomly chosen from a uniform distribution between \$50 to \$250 at specific time slots. For DQN, we have considered 8 hidden layers, each containing 64 nodes. We have considered the ReLU activation function for the hidden layer nodes and the output layer nodes use the softmax activation function. Graph data points are plotted based on the average results obtained from 50 simulation runs. Table 4 summarizes the performance parameter values and their corresponding ranges. All simulation evaluations were performed on a desktop computer equipped with an Intel(R) Core(TM) i5-8265U CPU @ 1.60 GHz 1.80 GHz, 8 GB memory, and Windows 10 (64 bit) Operating System.

5.2. Performance metrics

We have used the following standard performance metrics to compare the performance of the proposed OEAP-DQL system:

- **Quality of Experience (QoE):** It refers to the satisfaction or enjoyment that the electricity is received by the customers as per the demand for the electricity, which ranges from 0 to 1. The higher value indicates that the system can provide the maximum amount of electricity as per the demand of consumers.
- **Normalized Consumer Cost:** Consumer cost is defined as the required cost of the consumer to take the electricity service from SGOs. Normalized consumer cost can be determined by the ratio of consumer cost and budget, which is bounded in the range from 0 to 1. A higher value indicates a lower system performance.

- **Consumer Satisfaction:** Consumer satisfaction represents an integrated indicator in terms of the consumer's normalized cost and QoE. Consumer satisfaction, denoted as ξ and can be calculated as follows,

$$\xi = e^{-\frac{\hat{C}}{Q}}, \quad (26)$$

where, Q and $\hat{C}$ denote the average QoE of each consumer and the normalized consumer cost, respectively. And also It is bounded at the range from 0 to 1. This metric indicates the overall performance of the proposed system. When the consumer QoE is high and the normalized consumer power cost is low, overall customer satisfaction is higher and vice versa.

- **SGO Profit:** SGO earns the expected profit when the consumer's power demand is almost similar to their prediction, as the selling price is set according to the forecasted power demand. The total SGO profit at time t can be measured as follows,

$$P_{SGO}(t) = \sum_{i \in U} \{a_i(t) \times Y(t)\} - \sum_{j \in M} \{E_{d,j}(t) \times p_j(t)\}, \quad (27)$$

where, $Y(t)$ is the per unit selling price of power to consumer and $p_j(t)$ is the per unit purchase cost from EP j at time slot t ; $a_i(t)$ indicates allocated power to consumer $i \in U$ at time slot t and $E_{d,j}(t)$ is the predicted electricity demand that is purchased power from EP $j \in M$ at time slot t .

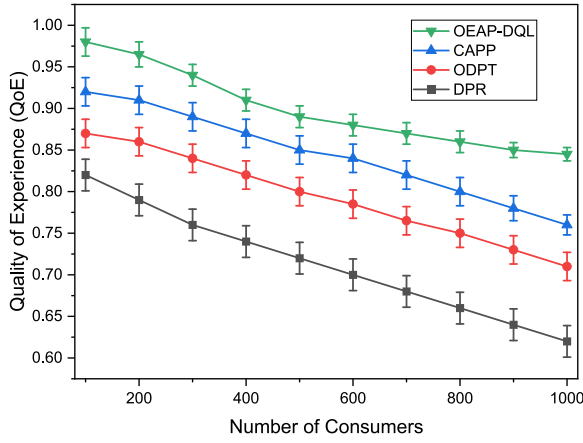
5.3. Result analysis

In this section, we characterize the effectiveness of the proposed OEAP-DQL system by comparing the results with existing models by varying the number of consumers, per unit selling prices, and different time slots.

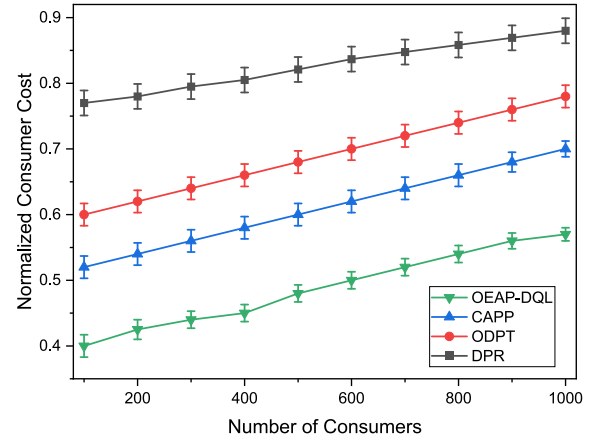
5.3.1. Impact of a varying number of consumers

In this experiment, we have varied the number of consumers of the overall system in the ranges of 100 to 1000. As an integral part of our analysis, we have also varied the selling price set by SGOs, which ranges from \$6 to \$16 based on the real-time pricing (RTP) schemes which depend on the level of consumer participation.

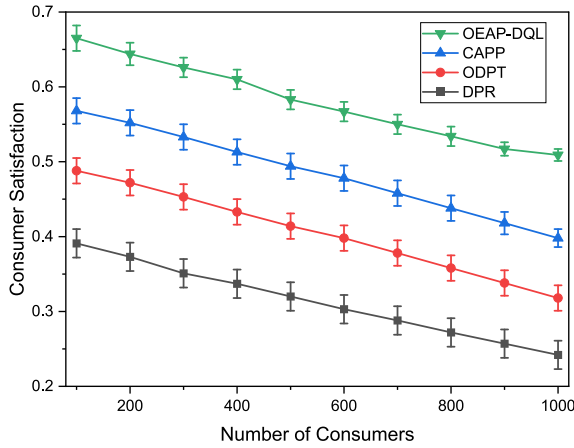
Fig. 6(a) shows that with the increasing number of consumers, the average QoE is decreased. QoE is lower in the DPR model because they have considered only trying to minimize consumer cost; however, they have not considered the QoE of each consumer. Although the CAPP, ODPT, and the proposed OEAP-DQL system perform closer to each other. However, OEAP-DQL performs slightly better than CAPP and ODPT. The ODPT primarily emphasizes maximizing the profit of the SGO while overlooking the impact on consumer costs. Their methodology relies on an Artificial Neural Network (ANN) based prediction model, which fails to incorporate essential external factors such as weather conditions and economic fluctuations, which in turn compromise the accuracy of their predictions. While CAPP aims to minimize consumer power costs, it falls short in prioritizing the enhancement of consumer QoE due to the absence of an electricity demand prediction model. Consequently, with the increasing number of consumers both CAPP and ODPT encounter challenges in effectively managing power distribution, leading to a decline in consumer QoE. On the contrary, the OEAP-DQL utilizes an advanced power demand forecasting model using the DQN technique to allocate power effectively, even with a growing



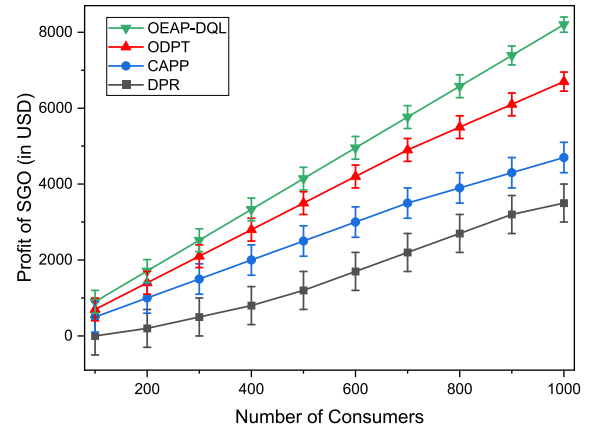
(a) Quality of Experience (QoE)



(b) Normalized Consumer Cost



(c) Consumer's Satisfaction



(d) Profit of SGO

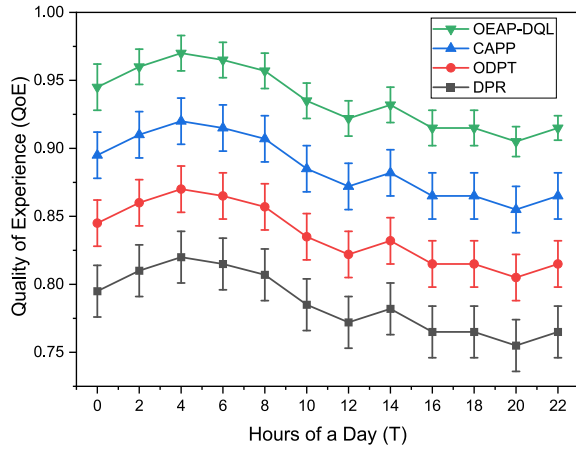
Fig. 6. Impacts of the varying number of consumers.

number of consumers. Consequently, OEAP-DQL outperforms DPR and ODPT in terms of QoE.

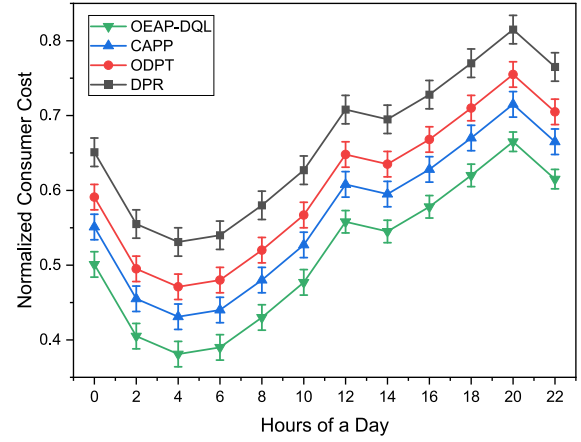
The increase in the number of consumers is directly linked to a rise in normalized consumer cost, as evidenced in Fig. 6(b). As the number of consumers increases, the power demand also increases, which can lead to high power costs for consumers. It is worth noting that the cost associated with the DPR approach significantly increases due to the implementation of a static Time of Use (ToU) pricing policy that spans over an extended duration, such as six months or one year. Additionally, the consumer power cost in the ODPT method surpasses that of the OEAP-DQL system since ODPT prioritizes the profit of the SGO rather than the consumer's interests, which in turn increases the consumer's cost in certain instances. OEAP-DQL and CAPP demonstrate comparable performance. While CAPP focuses on minimizing pricing policies related to user load profiles and residential demand responses, it emphasizes fair price distribution but lacks real-time adjustments for electricity demand and supply fluctuations. On the other hand, the proposed OEAP-DQL system incorporates three distinct pricing criteria based on the Real-time Pricing (RTP) policy, namely high-peak, mid-peak, and low-peak pricing schemes. This model ensures that when consumer participation is low, it generates lower power costs, while the number of consumers increases, indicating a high peak scheme with slightly higher power costs but an overall lower normalized consumer cost compared to other models. Consequently, the OEAP-DQL system demonstrates superior performance in terms of achieving reduced normalized consumer power costs.

Based on the findings depicted in Fig. 6(c), there is a noticeable decrease in overall consumer satisfaction with the growing number of consumers. The OEAP-DQL, CAPP, ODPT, and DPR systems demonstrate higher levels of satisfaction when consumer participation is low. The CAPP system employs optimization algorithms and load management to minimize consumer costs, leading to lower electricity bills and increased consumer satisfaction. Conversely, the ODPT focuses on maximizing the SGO profit and enhancing consumer utility without considering the associated consumer power costs. Furthermore, the DPR model solely concentrates on minimizing consumer costs through the implementation of ToU and fixed demand response strategies, which lessen the satisfaction rate of the consumers compared to other models. However, the OEAP-DQL surpasses the other methods in terms of customer satisfaction as the OEAP-DQL system prioritizes the minimization of consumer costs while simultaneously enhancing the QoE for consumers. Consequently, the OEAP-DQL system emerges as the superior method for achieving higher levels of consumer satisfaction.

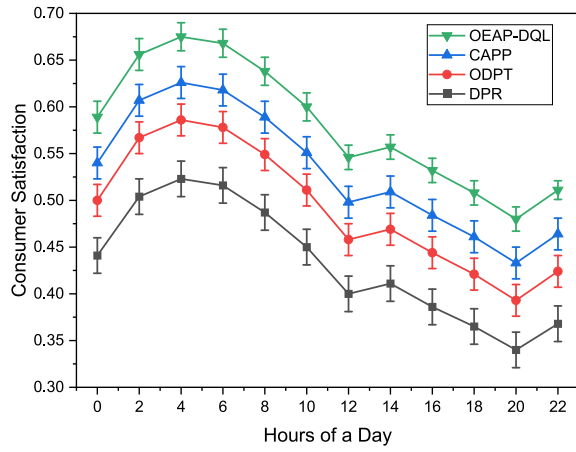
Fig. 6(d) illustrates that an increase in the number of consumers availing themselves of services provided by the service provider generally leads to higher profits for the provider. This positive correlation can be attributed to the fact that a larger consumer base typically generates more revenue, thus enhancing the provider's financial performance. The OEAP-DQL and ODPT methods exhibit almost similar performances in this regard. In the case of ODPT, the utilization of an ANN model enhances the accuracy of power usage prediction, consequently contributing to increased profits for the SGO. However, the authors of



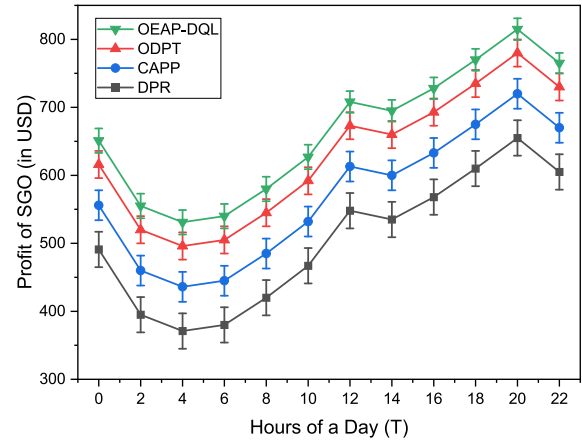
(a) Quality of Experience (QoE)



(b) Normalized Consumer Cost



(c) Consumer's Satisfaction



(d) Profit of SGO

Fig. 7. Impact of varying different time slot (T).

ODPT have not adequately considered the impact of external factors, such as weather conditions, and economic circumstances on power consumption. Conversely, in the OEAP-DQL system, we have established a prediction model based on the DQL approach, taking into account both internal and external constraints, which enables the prediction of future consumer demand accurately, ultimately facilitating increased operator profits. On the other hand, the DPR model employs a static ToU pricing policy that remains unchanged over an extended duration. The CAPP system emphasizes cost reduction for consumers, potentially limiting profit margins for SGOs striving for fair and nondiscriminatory electricity pricing, which might lead to decreased profits. Consequently, when the number of consumers increases, the SGO does not gain significantly higher profits compared to other models.

5.3.2. Impact of varying different time slot

In this experiment, we vary the time slots and then observe the overall system. The whole day is divided into 24 time slots, and the length of each slot is 1 h. We have varied the number of consumers in the range of 100 to 1000 in different time slots and the selling price of SGOs is varied between \$6 to \$16, which depends on the level of consumer participation.

Fig. 7(a) demonstrates a relationship between average QoE and different time slots of a day. The non-linear relationship between these two metrics is because of the number of participation of the

consumers and fluctuations in the demand. Our model utilizes a DQL-based prediction model and divides the 24 time slots into low-peak, mid-peak, and high-peak periods. We have considered 1 AM to 8 AM to be the low-peak region, 9 AM to 10 AM, and 1 PM to 2 PM to be the mid-peak region, and the rest of them are in the high-peak region. Additionally, we incorporate power storage to support our model during uncertain power demand scenarios. This enables service providers to effectively satisfy consumer demand and achieve higher QoE. In comparison, the DPR model's fixed prediction model leads to lower QoE by not considering individual consumer QoE or adequately mitigating high consumer power demand. On the other hand, the proposed OEAP-DQL system performs similarly to the CAPP and ODPT but with slight improvements. The ODPT relies on an ANN-based prediction model that lacks consideration for external factors, resulting in less accurate predictions and prioritization of SGO profit over consumer QoE. On the other hand, due to the absence of a demand prediction model in CAPP, it limits its ability to provide uninterrupted electricity according to the demand, impacting consumer QoE. However, its focus on reducing consumer power costs hints at a potential enhancement in QoE compared to ODPT, not OEAP-DQL. As a result, consumer satisfaction is compromised, highlighting the superior QoE performance of the OEAP-DQL system.

Fig. 7(b) depicts the variation in consumer power cost across different time slots, influenced by their power demand. As the number of consumers increases, leading to high-peak periods, resulting in higher

power costs for consumers. Conversely, when consumer power demand decreases, costs decrease as well. In the OEAP-DQL system, we introduce three distinct pricing sets for low-peak, mid-peak, and high-peak periods and the effective prediction exploiting the DQN model enables us to lower the consumer power costs compared to other models. Likewise, CAPP achieves low consumer power costs by emphasizing cost minimization through a fair pricing policy. However, it lacks the integration of a prediction model and dynamic pricing schemes. Conversely, in the DPR model, costs are high due to the adoption of a fixed pricing policy for an extended duration. The DPR model fails to consider consumer participation, leading to inflexible charges regardless of power demand fluctuations. Similarly, in ODPT, consumer power costs are high as the focus remains solely on the SGO's profit, with an underperforming prediction model and a lack of notable pricing schemes. In contrast, the OEAP-DQL system excels in minimizing consumer power costs, establishing its superiority in terms of normalized consumer power expenses.

In Fig. 7(c), we observe that consumer satisfaction varies across different time slots based on their power demand. The non-linear relationship between consumer satisfaction and different time slots is characterized by the fact of dynamic participation, power demand of the consumers, and random events. Consumer satisfaction increases significantly when power demand decreases and vice versa. In terms of the specific parameters used to gauge consumer satisfaction, the proposed OEAP-DQL system outperforms other methods. Our model focuses on minimizing consumer costs while simultaneously enhancing their QoE. In this regard, the CAPP extensively uses optimization algorithms to reduce consumer costs, resulting in lower electricity bills. However, the lack of emphasis on consumer QoE contributes to lower consumer satisfaction compared to OEAP-DQL. In contrast, the ODPT method prioritizes maximizing the SGO's profit and enhancing consumer utility, neglecting the associated consumer power costs. On the other hand, the DPR method solely concentrates on minimizing consumer costs through fixed demand response strategies and ToU pricing, which in turn exhibits significantly lower consumer satisfaction compared to other models. As a result, the OEAP-DQL method emerges as the superior approach for achieving higher levels of consumer satisfaction.

Fig. 7(d) illustrates the impact of consumer power demand on the SGO profit across different time slots. In this regard, both the OEAP-DQL and ODPT demonstrate comparable performance. The ODPT benefits from the utilization of an ANN-based prediction model, enhancing the accuracy of power usage prediction and resulting in increased profits for the SGO. However, external factors like weather conditions and economic circumstances have been overlooked, which influence power consumption. In contrast, the OEAP-DQL system incorporates a comprehensive approach by employing a Deep Q-Learning prediction model, which can accurately predict future consumer power demand and considers both internal and external constraints. Additionally, the integration of RTP schemes into the OEAP-DQL system leads to higher profits for the SGO. In this comparison, CAPP prioritizes cost reduction for consumers through fair pricing, potentially impacting SGO profits negatively as their earnings are not the primary focus. On the other hand, the DPR model adopts a static ToU pricing policy that remains unchanged for extended periods. As a result, when the number of consumers increases, the SGO does not experience significant profit gains compared to other models. Therefore, OEAP-DQL outperforms other methods in terms of maximizing SGO's profits.

5.3.3. Impact of the varying selling price of SGO

In this experiment, we have varied the selling price of the SGOs within the range of \$6 to \$16 to observe the overall system performance, where we have fixed the number of consumers at 350.

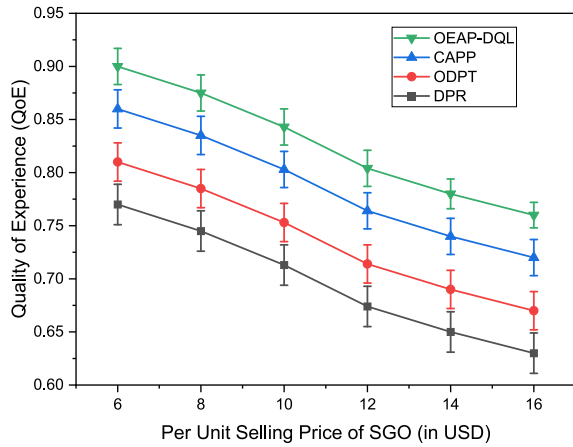
Fig. 8(a) illustrates that the increased selling price of SGO leads to a decrease in the average consumer QoE. Economically, when the selling price increases, consumers fail to get the required service according to

their desired level and this may lead to disappointment or dissatisfaction of the consumer, which in turn lessen QoE. The OEAP-DQL system exploits an advanced forecasting system, which combines advanced power demand forecasting techniques with DQL to optimize power allocation effectively. Despite the increase in selling price, OEAP-DQL outperforms other methods in terms of QoE, providing customers with a more satisfactory experience. However, due to the lack of a predictive model, CAPP may face suboptimal power allocation in certain scenarios, adversely affecting consumer QoE. Yet, its emphasis on lowering consumer power costs suggests a potential QoE improvement compared to ODPT. On the other hand, the ODPT method relies on an ANN-based prediction model, which does not take into account external factors such as weather, and economy, resulting in less accurate predictions. As a result, power allocation may not be optimal in certain situations. For that reason, the OEAP-DQL system outperforms other methods in terms of QoE for accurately predicting the power demand.

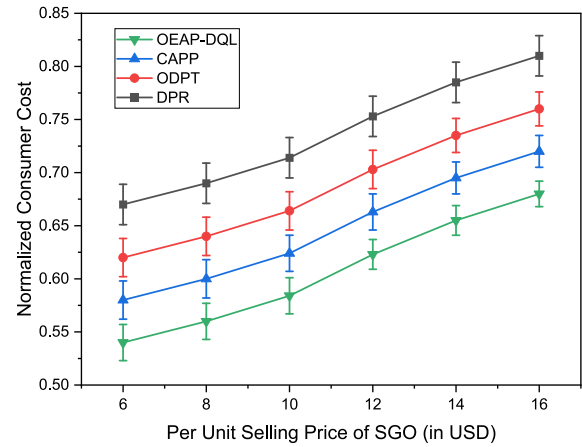
According to Fig. 8(b), increasing the selling price of SGO leads to an increase in the normalized consumer cost. When the selling price of SGO rises, consumers are required to make a higher initial payment to obtain or utilize the product, directly impacting their upfront expenses. Consequently, the power cost for consumers also increases as a result of the increased selling price of SGO. In the OEAP-DQL, our approach takes into account real-time pricing based on the consumer's power demand, which is updated on an hourly basis. This means that the consumer's power cost fluctuates hourly, depending on their power demand within the SGO. As a result, OEAP-DQL yields better results compared to CAPP, ODPT, and DPR. In contrast, the CAPP primarily focuses on reducing consumer costs through a fair pricing policy. However, this policy lacks exploitation of RTP policy similar to OEAP-DQL. ODPT primarily focuses on maximizing SGO's profit and does not consider any parameters related to consumer cost. On the other hand, DPR follows a fixed ToU pricing policy over an extended period, resulting in significantly higher consumer power costs. Consequently, the OEAP-DQL outperforms the other methods in terms of normalized consumer cost.

Fig. 8(c) illustrates a significant decline in overall consumer satisfaction when the selling price of SGO is increased. Among the methods analyzed, namely, OEAP-DQL, CAPP, ODPT, and DPR, higher consumer satisfaction is observed when the selling price of SGO remains low. However, when considering the specific parameters related to consumer satisfaction, OEAP-DQL outperforms the other methods. Our model places a strong emphasis on minimizing consumer costs while simultaneously enhancing consumer QoE. In this aspect, the CAPP model employs optimization algorithms for consumer cost reduction without focusing on consumer QoE, which results in decreased consumer satisfaction. On the other hand, the ODPT focuses on maximizing SGO's profit and improving consumer utility without adequately considering the associated consumer power costs. Additionally, the DPR primarily focuses on minimizing consumer costs through the implementation of ToU and fixed demand response strategies, which in turn significantly lessens consumer satisfaction compared to the other models. As a result, OEAP-DQL emerges as the superior method for achieving higher levels of consumer satisfaction in our research.

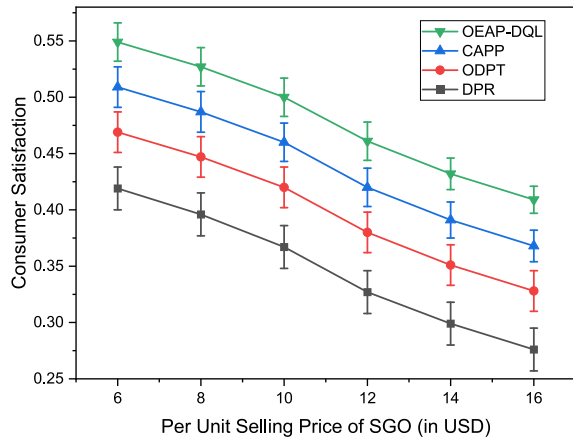
In Fig. 8(d), it is evident that increasing the selling price set by the SGO leads to an increase in SGO profit. This aligns with economic principles where higher prices result in higher profits for service providers. In our analysis, we have found that \$12 is a significant price point for consumers because, with the increased selling price after \$12, consumers will not purchase the quantity of electricity will not increase significantly, which lessens the profit of the SGOs. Both OEAP-DQL and ODPT exhibit almost similar performance in terms of SGO profit. However, in the proposed OEAP-DQL method, we incorporate power storage capabilities to store excess power. This additional power can be sold to consumers at any time, eliminating power wastage and contributing to higher SGO profits even with a higher selling price. In contrast, ODPT, CAPP, and DPR models do not account for power



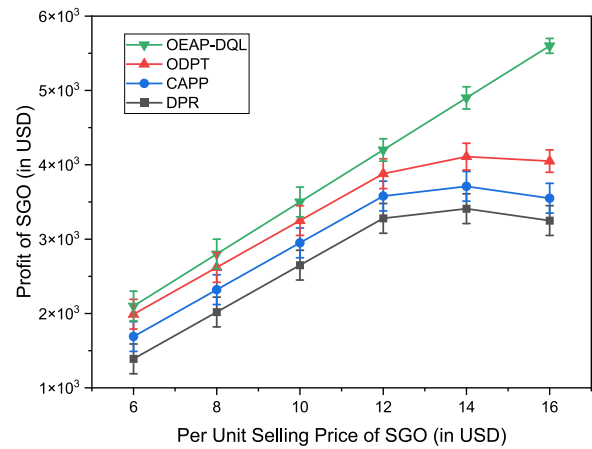
(a) Quality of Experience (QoE)



(b) Normalized Consumer Cost



(c) Consumer's Satisfaction



(d) Profit of SGO

Fig. 8. Impact of varying per unit selling price.

storage, which means that if they increase the selling price, consumer demand decreases and results in power wastage. Consequently, OEAP-DQL outperforms other methods by effectively managing power storage and achieving higher SGO profits even with a higher selling price.

5.3.4. Prediction accuracy

Fig. 9 illustrates that the prediction accuracy of the proposed OEAP-DQL system is better results compared to other prediction models. It chooses the best prediction model from multiple options of actions at each time slot t . Compared to the conventional EWMA, ARIMA, as well as SARIMA models, the suggested OEAP-DQL system significantly improves the accuracy of electricity demand forecasts. Concerning the prediction error ratio, OEAP-DQL demonstrates superior efficiency. The error rates for EWMA, ARIMA, and SARIMA are 3.4%, 3.7%, and 2.1%, respectively, while OEAP-DQL achieves a notably lower error rate of 1.01%. Here, the OEAP-DQL system takes advantage of different behaviors during different times of the day or states learned during mid-peak, low-peak, and high-peak energy demands. Conversely, EWMA, as well as ARIMA models do not utilize multiple action types when calculating the energy demand forecast. As a result, the OEAP-DQL system outperforms the other models in terms of prediction accuracy.

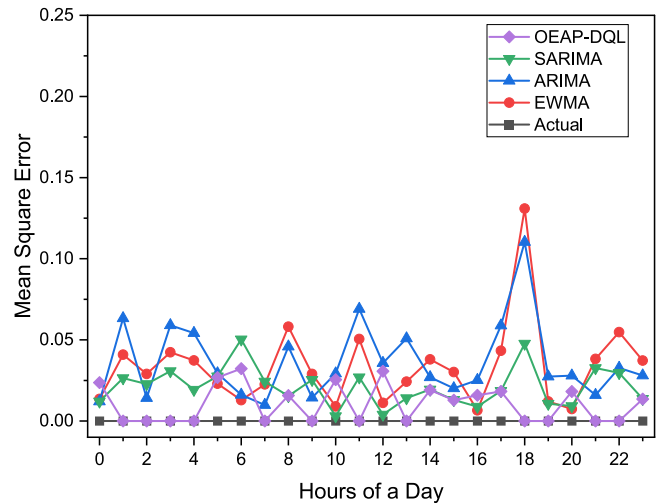


Fig. 9. Forecasting error of EWMA, ARIMA, SARIMA, Actual, and OEAP-DQL.

6. Conclusion

In this paper, we developed a framework namely OEAP-DQL that brought a trade-off between maximizing consumers QoE and minimizing consumer cost. However, due to the fluctuations of the electric power demand, accurate prediction of the electricity demand in future time slots is a critical factor in enhancing the consumer's QoE and minimizing cost. For that reason, we also presented a method for forecasting future electricity demand in a smart grid system exploiting Deep Q-Learning. Due to the incorporation of a DQL-based electricity demand forecasting model and real-time pricing mechanism, the developed OEAP-DQL system maximized the consumer's QoE and minimized cost, which in turn enhanced consumer satisfaction. The experimental results carried out in Python, demonstrated significant improvements over state-of-the-art works, in terms of consumer QoE and power cost up to 10% and 15%, respectively. In the future, we plan to develop mathematical models that can account for factors such as critical peak demand and uncertainties caused by consumers and machine failures, to achieve even better performance.

CRedit authorship contribution statement

Abdur Razzak: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft. **Md. Tariqul Islam:** Conceptualization, Data curation, Investigation, Methodology, Software, Writing – original draft. **Palash Roy:** Data curation, Funding acquisition, Investigation, Methodology, Resources, Software, Writing – review & editing. **Md. Abdur Razzaque:** Conceptualization, Funding acquisition, Investigation, Project administration, Supervision, Validation, Writing – review & editing. **Md. Rafiul Hassan:** Investigation, Validation, Visualization, Writing – review & editing. **Mohammad Mehedi Hassan:** Funding acquisition, Resources, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

We would like to declare that the authors do not have any technical conflict of interest in the submission of the journal paper and its publication in the ENERGY journal of Elsevier. We also declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors did not use any generative AI or AI-assisted technologies.

Acknowledgments

This work was supported by the King Saud University, Riyadh, Saudi Arabia, through the Researchers Supporting Project under Grant RSP2024R18. This work was also supported in part by the Center for Research, Innovation, and Transformation (CRIT) of Green University of Bangladesh under Grant G-23-11-09.

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